

----- Active space -----

RICKETS AND SCURVY

khanirfan0392@gmail.com

Rickets

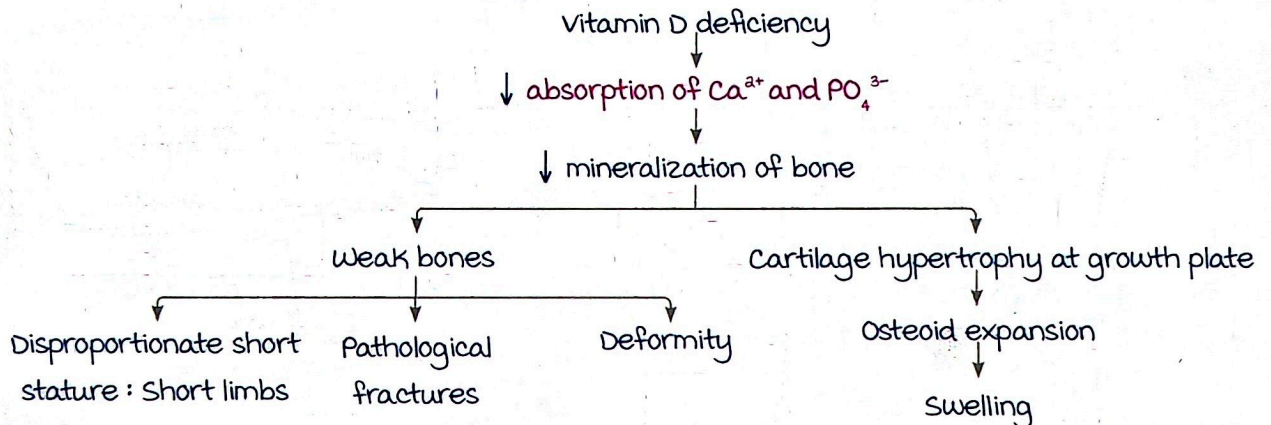
00:00:27

Occurs due to vitamin D deficiency (Sunshine vitamin).

ETIOLOGY :

- Dietary deficiency (Fish, fish liver oil, egg yolk).
- Drugs :
 - Anti-epileptics : Phenytoin, phenobarbitone, carbamazepine.
 - Antitubercular drugs : Isoniazid and rifampicin.
- ↓ sunlight exposure.
- malabsorption.
- Chronic liver disease.

PATHOPHYSIOLOGY :



DEFORMITY :

Head :

1. Craniotabes :
 - Softening of skull bone.
 - Sponge ball/ping pong ball like consistency.
2. ↑ size of fontanelles (Wide open).
3. Frontal bone bossing.

Note :

Craniotabes can also be seen in :

- Congenital syphilis.
- Hydrocephalus.
- Osteogenesis imperfecta.

Chest :

1. Costochondral junction widening :

Rachitic rosary

(Painless & round).

- AKA string of beads appearance.

Costochondral junction widening

Harrison's sulcus



Rickets chest

Paediatrics • v1.0 • Marrow 8.0 • 2024

Page 1/5

2. Harrison's sulcus groove : D/t pull of diaphragm on weakened ribs.

----- Active space -----

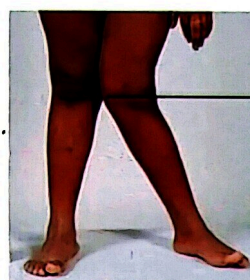
Limbs :



1. Widening of wrist



2. Genu varum/bow leg



3. Genu valgus/knock knees

4. Double malleolus sign : Cartilage hypertrophy (Swelling) above lateral malleolus.

Other findings :

1. Delayed dentition.
2. Seizures or tetany : In infants d/t acute hypocalcemia.

INVESTIGATIONS :

Blood investigations :

- ↓ Vitamin D : ↓ 25 HCC.
- D/t ↓ absorption :
 - ↓ S. Ca^{2+} → ↑ PTH → Normal S. Ca^{2+} .
 - ↓ S. PO_4^{3-} .
- D/t ↑ bone resorption by PTH : ↑ ALP.

Note : Classification based on Vit D levels Serum Vit D levels.

	Serum vitamin D levels (ng/ml)
Sufficiency	> 20
Insufficiency	12 - 20
Deficiency	< 12

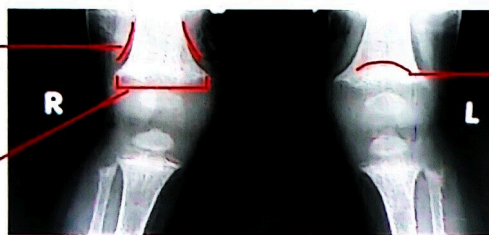
Radiological investigation :

X-Ray : Primary tool of diagnosis.

- Loss of zone of provisional calcification (ZPC) : white line at end of long bone.

Splaying : Lateral deviation of ends

Fraying (First sign) : Irregularity at ends



Cupping

Rickets

----- Active space -----

TREATMENT :

1. Daily dose of vitamin D → Recommended.
- Duration : 12 weeks.

Age	Dose of vit D
< 1 year	2000 IU/day
> 1 year	3000 IU/day

2. Ca^{2+} supplement : 50 - 75 mg/kg/day.

monitoring response :

1. X-Ray : Preferred method.

Shows changes by 4 weeks.

↓
Healing with line of rickets (1st visible sign)

2. Biochemical investigation : Shows changes in 12 weeks.

Refractory Rickets

00:18:52

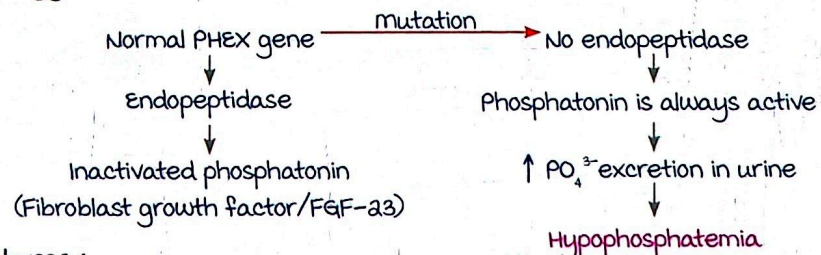
No response to vitamin D supplementation.

DISORDERS SEEN :

1. Familial hypophosphatemic rickets.
2. Vitamin D dependent rickets (VDDR).
3. RTA associated rickets.
4. CKD associated rickets.

Familial hypophosphatemic rickets :

- X-linked dominant inheritance.
- *m/c inherited form* of rickets.
- **PHEX gene mutation** (Codes for phosphate regulation endopeptidase).

Pathophysiology :**Clinical features :**

- Lower limb deformities.
- **No features of hypocalcemia** (Eg : Tetany).
- Dental abnormalities : Pulp exposure, ↑ risk of dental abscess.

Rickets & scurvy

71

Treatment :

1. Phosphate supplementation : **Joulie solution** (Contains elemental phosphates).
2. Burosumab : Newer drug : monoclonal antibody against FGF-23.

Vitamin D dependent rickets (VDDR) :

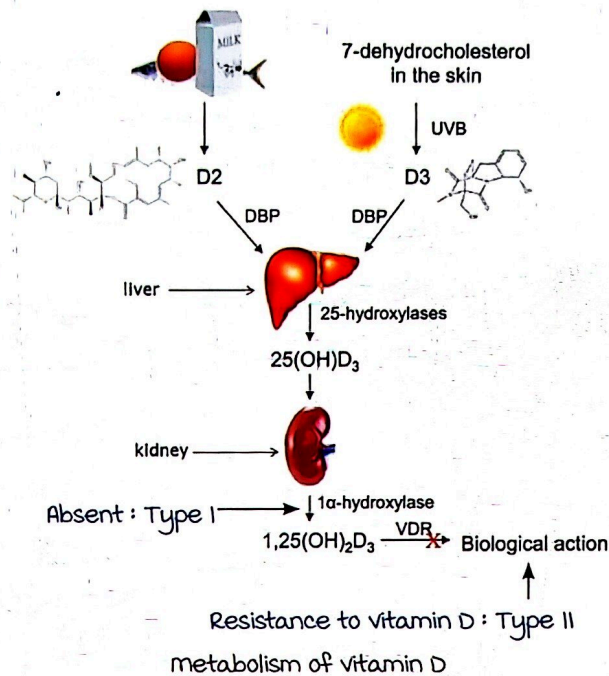
- Autosomal recessive inheritance.
- Defects in vitamin D metabolism $\rightarrow$ Hypocalcemia.

Clinical features :

1. Classic features of rickets.
2. Additional features in Type II VDDR :
 - Ectodermal defects : milia, epidermal cysts.
 - Oligodontia ($\downarrow$ teeth).
 - Alopecia.

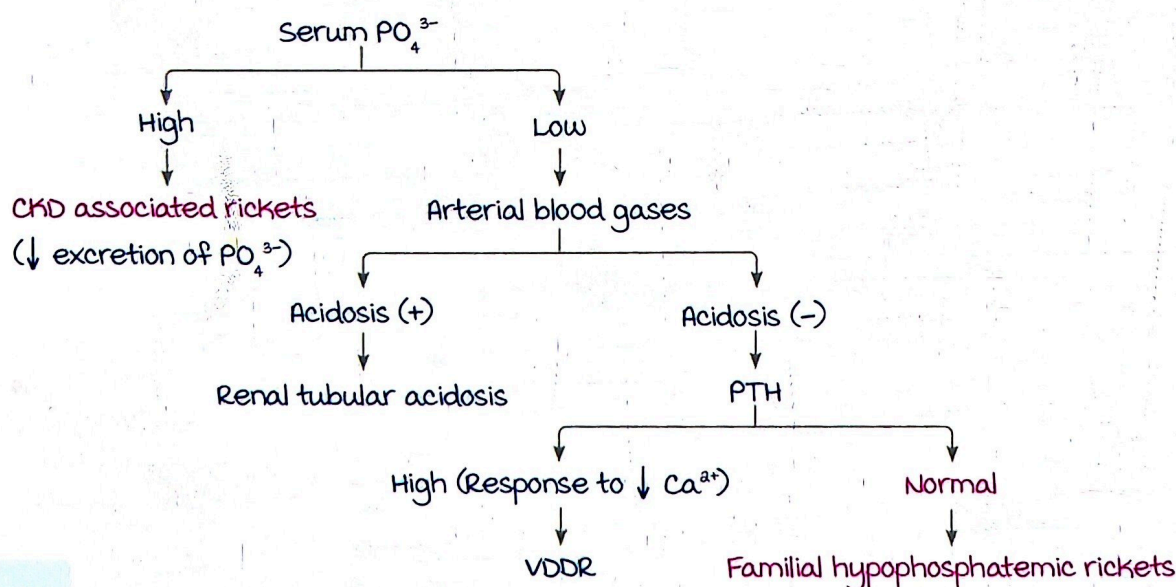
Treatment :

1. Type I :
 - α calcidol (Active form of vit D).
 - Ca^{2+} supplements.
2. Type II :
 - High doses of vit D $\rightarrow$ Partially overcome resistance $\rightarrow$ Improves symptoms.



APPROACH TO REFRACTORY RICKETS :

Based on biochemical values.



72

Nutrition

----- Active space -----

Scurvy

00:32:33

D/t **vitamin C deficiency.****ROLE OF VITAMIN C :**

1. Collagen formation

Proline and lysine

↓ Hydroxylation (Requires vitamin C)

mature collagen

2. Wound healing.

3. Osteoid/bony trabeculae formation.

ETIOLOGY :

1. Vitamin C deficient diet food rich in vit C :

- Citrus fruits : Lemon, orange
- Green vegetables

2. Intake of boiled/evaporated cow milk (Without fruit/vegetables).

CLINICAL FEATURES :

Early features :

- Abnormal irritability.
- Anemia : vit C is needed for iron absorption.
- ↓ appetite.

Bleeding manifestations : Characteristic.

- Gum bleeds.
- **Perifollicular bleeding** + corkscrew hair.
- Bone : Subperiosteal bleeding → **Pain (+)** → Pseudo paralysis

↓
Child keeps his hips & knees semiflexed.↓
Frog leg posture

Rosary :

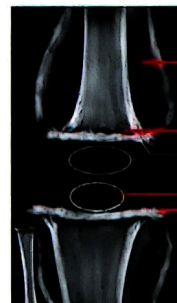
- Painful.
- Sharp or angulated.

DIAGNOSIS :mainly **clinical diagnosis.**

X-Ray :

TREATMENT :

- Oral vitamin C : 100 - 200 mg/day.
- Duration : Until lesion subsides.



Subperiosteal hemorrhage

Area of rarefaction (Poorly formed bony trabeculae)

Ring epiphysis

White line of Frankel

Central area :

Trummerfeld's zone

Lateral area :

Pelken's spur